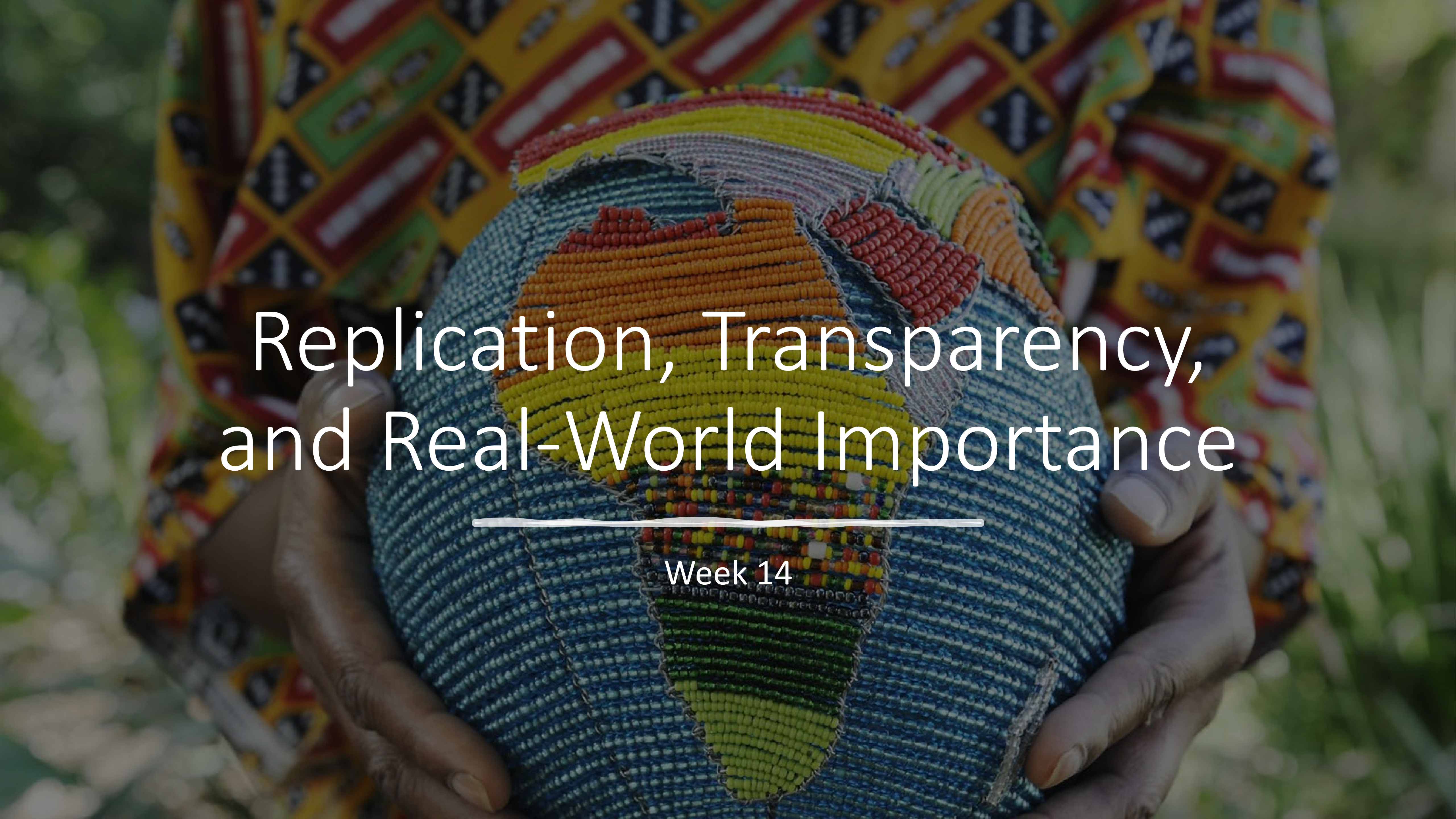


Replication, Transparency, and Real-World Importance

Week 14

Roadmap

Ethics

- Institutional Review Board (IRB)
- Ethical Issues
- Scientific Misconduct

Replication

- Types of replication
- Replication projects
- Meta-analysis: What does the literature say?
- Replicability and popular media

Research transparency and credibility

- Questionable research practices
- Transparent research practices

Must a study have external validity?

- Generalizing to other participants
- Generalizing to other settings
- Does a study have to be generalizable to many people?
- Does a study have to take place in a real-world setting?

Institutional Review Board (IRB)

Members are from both scientific and nonscientific disciplines.

- At least one member must be from the community.

Researchers submit a written proposal describing

- the purpose, procedures, and potential risks of the study.

IRB must approve the proposal

- before the study may be conducted.

Ethical Issues

- Lack of adequate informed consent
- Invasion of privacy
- Coercion to participate
- Potential physical or mental harm
- Deception
- Violation of confidentiality

Informed Consent

Inform participants of the nature of their participation in the study and obtain their explicit agreement to participate.

Problems with obtaining informed consent:

- Compromising the validity of the study
- Participants who are unable to give informed consent
- Ludicrous cases of informed consent

Elements of Informed Consent

- Description of why the study is being conducted
- List of activities that participant will do
- Description of possible risks
- Statement informing participants that they may refuse to participate with no penalty
- Confidentiality statement
- Encouragement for participants to ask questions
- Instructions on how to contact the researcher
- Signature lines for the researcher and the participant

Privacy

Participants have the right to decide

- when, where, to whom, and to
- what extent their responses will be revealed.

Research involving observation of people

- in public places does not constitute an invasion of privacy.

Coercion

Coercion to participate occurs when

- participants agree to participate in a research study because of
- real or implied pressure from some individual who has
- authority or influence over them.

When research participation is

- a course requirement or given as extra credit,
- students must be given the option of an alternate activity
  - for filling the requirement or earning credit.

Physical and Mental Stress

- The focus of a study may require participants to
  - experience pain, stress,
  - failure, anxiety, or
  - other negative emotions.
- Minimal risk –
  - risk that is no greater in probability and severity
  - than that ordinarily encountered in daily life or
    - during the performance of routine physical or
    - psychological examinations of tests
- Any use of more than minimal risk requires strong justification.

Deception

- Types:
  - Presenting participants with a false purpose of the study
  - Using an experimental confederate who poses as another participant or a bystander
  - Providing false feedback to participants
  - Presenting two related studies as unrelated
  - Giving incorrect information regarding stimulus materials

Confidentiality

Confidentiality – a participant's data may be used only for purposes of the research and may not be divulged to others

The easiest way to maintain confidentiality is to ensure that participants' responses are anonymous.

Scientific Misconduct

- Categories of scientific misconduct:
  - Fabrication, falsification, and plagiarism
  - Questionable research practices
  - Unethical behavior

Questionable research practices

- Assuming “ownership” over more than one’s contribution to the study
- Failing to report findings that are inconsistent with one’s views

Unethical behavior

- such as sexual harassment,
- abuse of power,
- discrimination, or
- failure to follow government regulations

Wrapping Up...

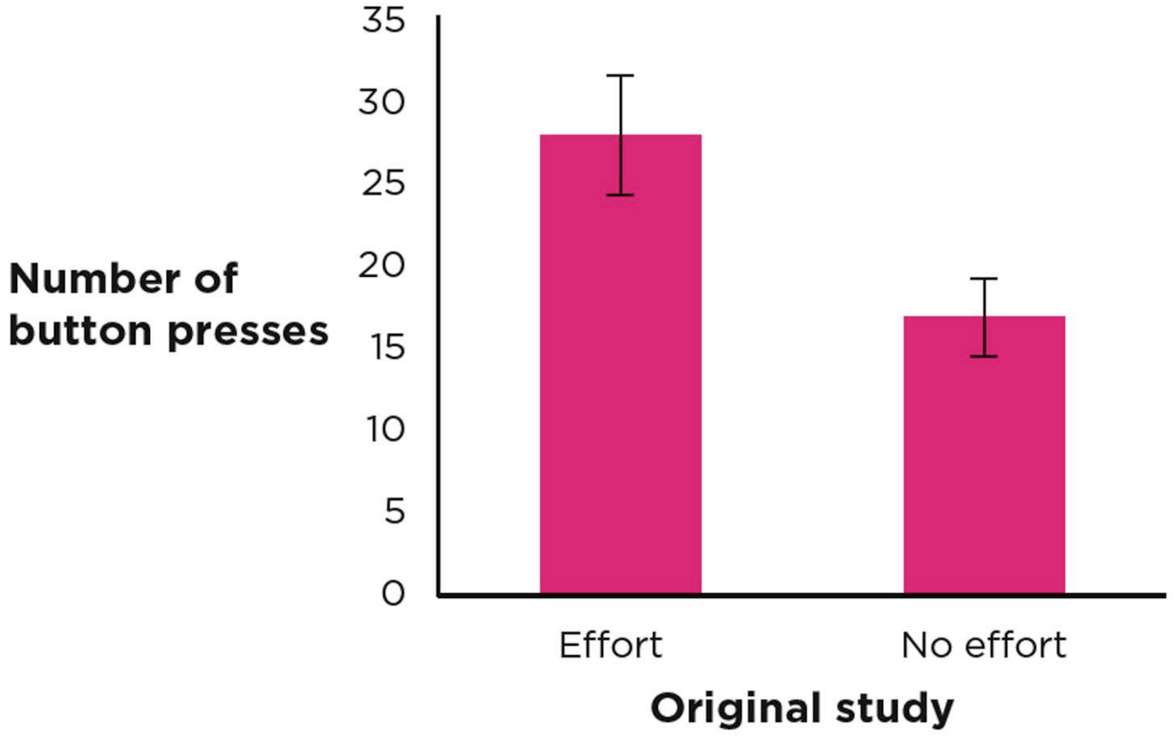
Replication

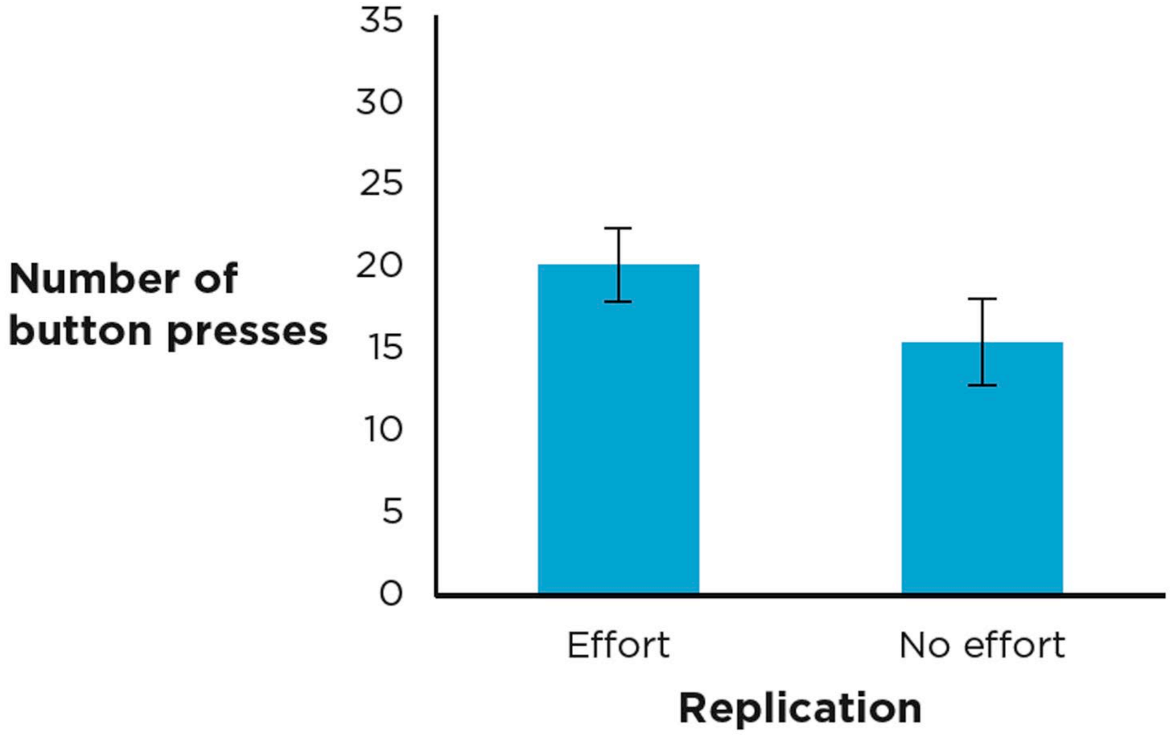
- A study that is replicable is reproducible;
  - not merely hypothetically
  - result has been repeated

Types of Replication

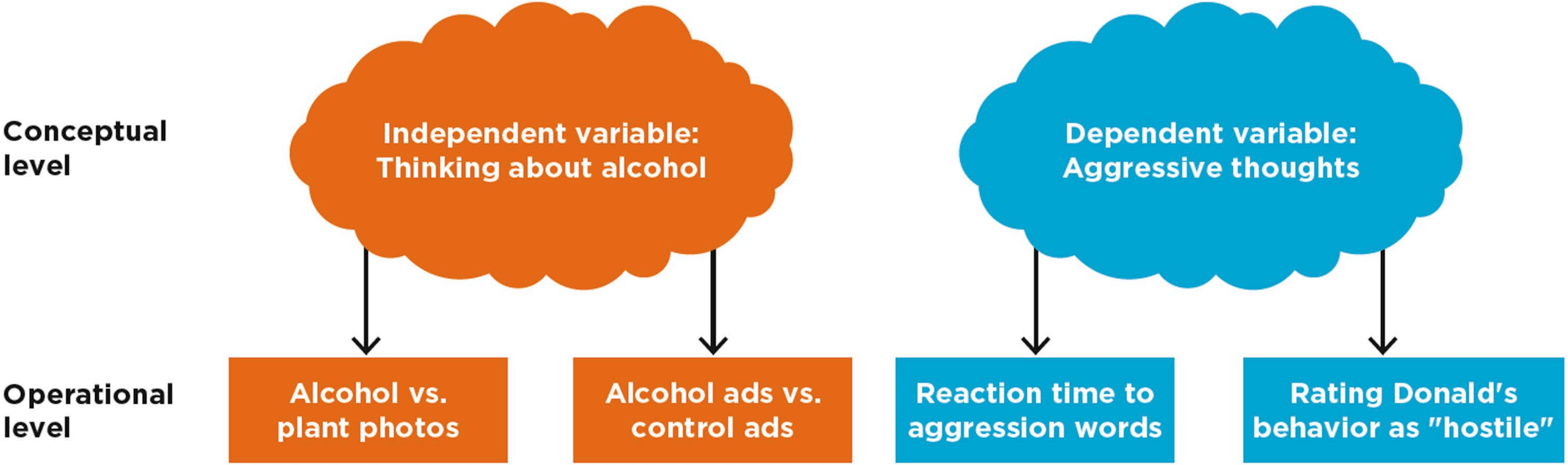
- Direct replication
- Conceptual replication
- Replication-plus-extension

Direct Replication

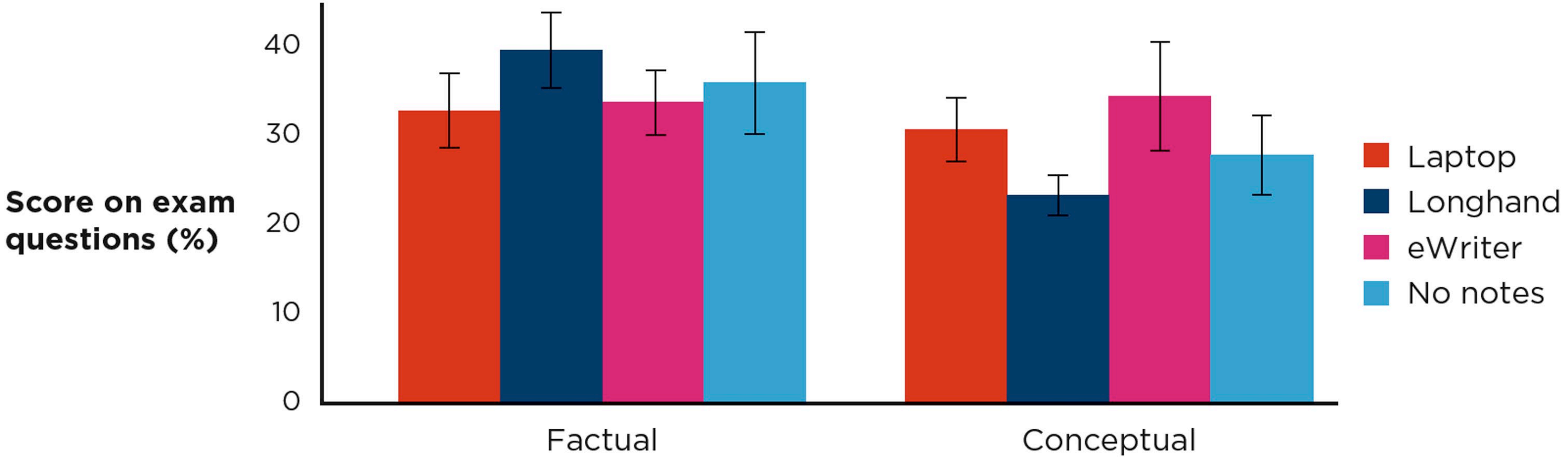




Conceptual Replication



Replication-Plus-Extension



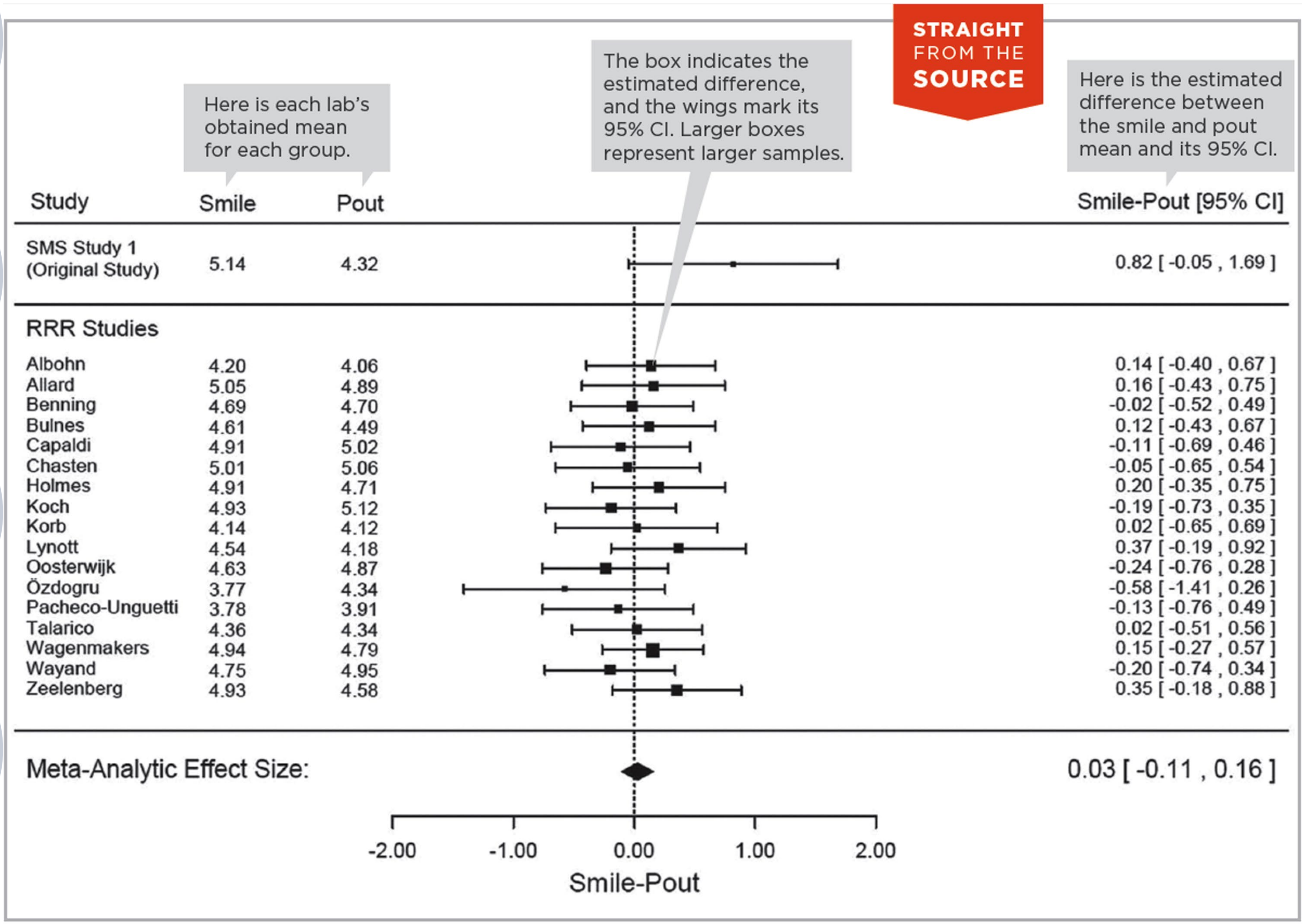
Why Might a Study Not Be Replicable?

- Contextually sensitive effects
- Number of replication attempts
- Problems with original study
  - Sample Size
  - Harking
  - P-Hacking

Improvements to Scientific Practice

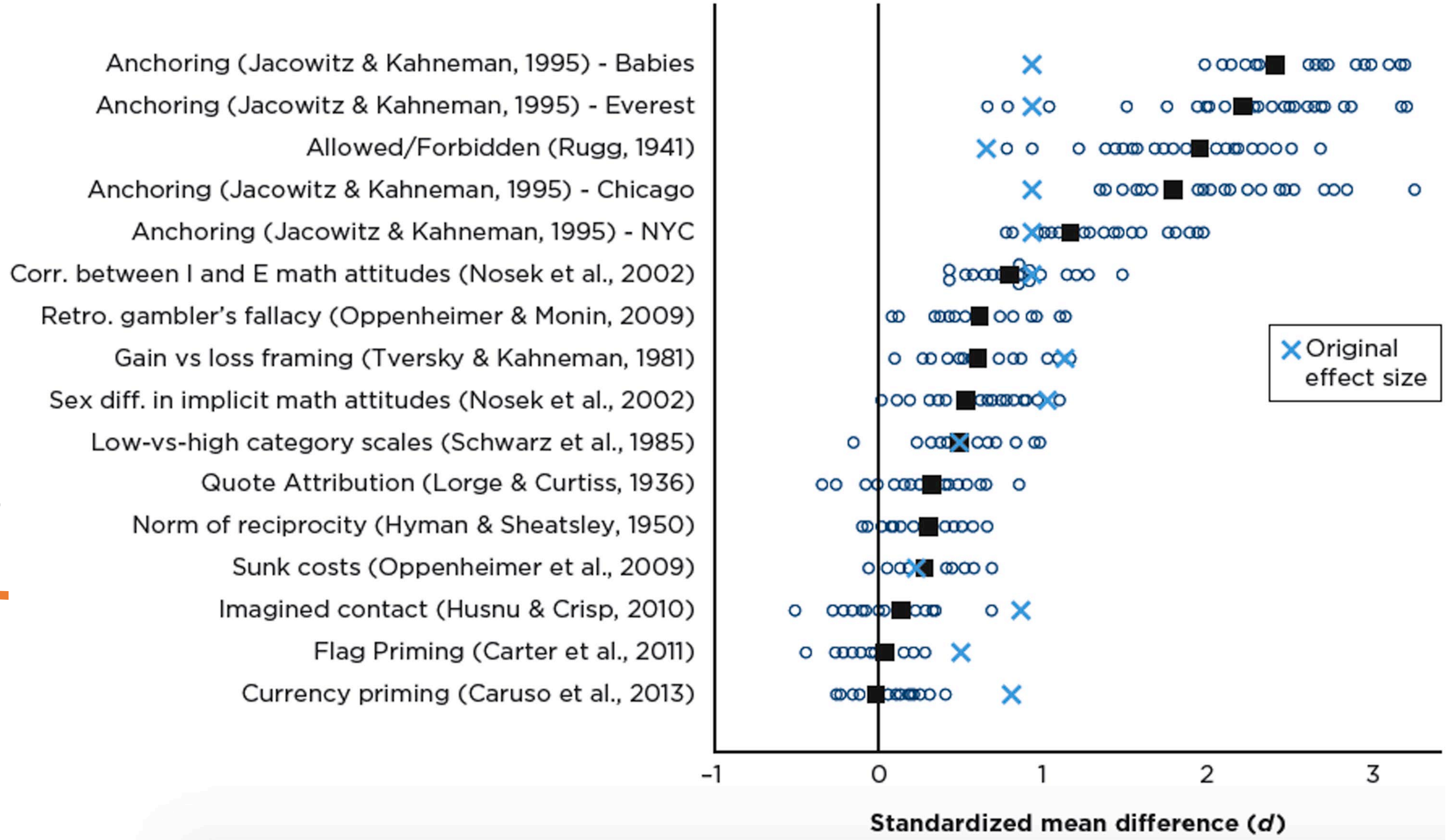
- Larger sample sizes
- Report all analyses and variable
- Open Science Collaboration
- Preregistration
- Replication Projects

One Study, Many Labs



Wagenmakers, E. J., Beek, T., Dijkhoff, L., Gronau, Q. F., Acosta, A., Adams, R. B., Jr., . . .
Zwaan, R. A. (2016). Registered Replication Report: Strack, Martin, & Stepper (1988). Perspectives on Psychological Science, 11, 917–928

Many Labs, Many Studies



Meta-Analysis: What Does the Literature Say?

Scientific literature (aka literature):

- a series of related studies conducted by different researchers who have tested similar variables.

Meta-analysis:

- a statistical analysis that yields a quantitative summary of a scientific literature.

Example: Lying takes time

Example: Lying Takes Time

TABLE 14.1

Results for the Reaction Time Difference Between Lying and Truthful Conditions, Categorized by Type of Instructions to Participants

TYPE OF STUDY	NUMBER OF STUDIES	TOTAL NUMBER OF PARTICIPANTS	AVERAGE EFFECT SIZE *d* (LYING CONDITION MINUS TRUTHFUL CONDITION) AND 95% CI
Participants instructed to go as fast as possible	100	2,866	1.29, [1.16, 1.41]
Participants not instructed to go fast	11	354	0.97, [0.64, 1.30]

TABLE 14.2

Results for the Reaction Time Difference Between Lying and Truthful Conditions, Categorized by Participants' Motivation to Avoid Detection

TYPE OF STUDY	NUMBER OF STUDIES	TOTAL NUMBER OF PARTICIPANTS	AVERAGE EFFECT SIZE *d* (LYING CONDITION MINUS TRUTHFUL CONDITION) AND 95% CI
No motivation	85	2,552	1.33, [1.19, 1.47]
Instructed to avoid detection	23	626	1.00, [0.78, 1.22]
Incentive to avoid detection	6	129	1.13, [0.84, 1.43]

Example: Lying Takes Time

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Strengths and Limitations of Meta-Analysis

File drawer problem

- Null results and opposite results rarely published

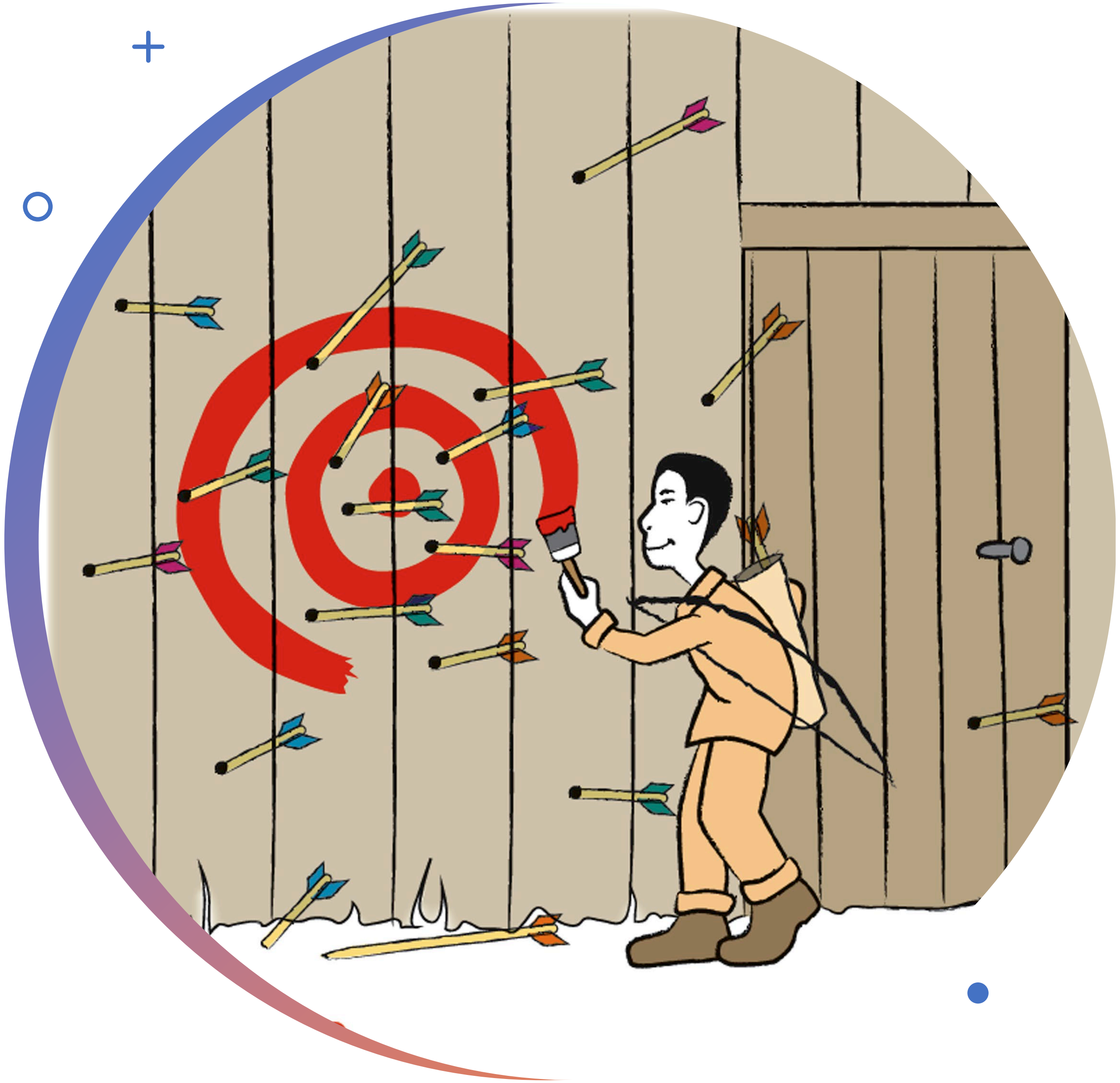
Example: Antidepressants and depressive symptoms

Replicability and Popular Media

Journalists do not always consider the importance of replicability.

A responsible journalist will not only report on the latest and most exciting studies but also summarize the body of literature on the topic.

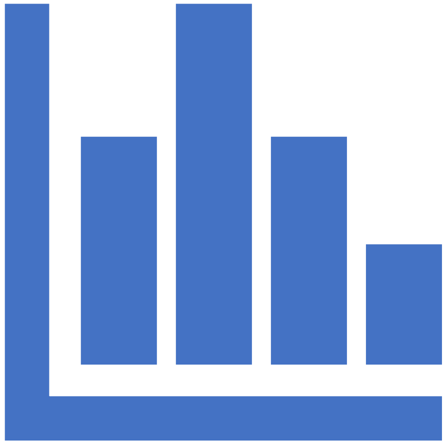
Wrapping Up…



Research Transparency and Credibility

- Questionable research practices
  - Underreporting null effects
  - p-Hacking
  - HARKing
  - Using Small Samples
- Transparent research practices

Underreporting null effects



- Researchers mislead about the strength of the evidence
  - by not reporting conditions or measures that did not support the hypothesis.
- Solution?
  - Open materials, in which all study materials are reported publicly
  - Others can see the full study design and fairly evaluate the strength and consistency of the evidence.

p-Hacking

Researchers try many ways of analyzing their data, so the result is more likely to be a fluke rather than a true, replicable pattern.

Solution?

- Open data, in which full data sets are provided
- Others can re-run and confirm the statistical analyses.
- They can also use the data to test new questions.

Underreporting and *p*-Hacking

TABLE 14.3

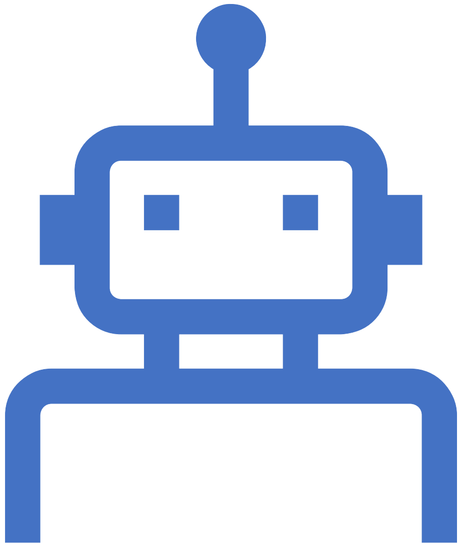
Questionable Versus Transparent Research Practices

QUESTIONABLE PRACTICE	EXPLANATION	ALTERNATIVE, TRANSPARENT PRACTICE	EXPLANATION	BADGE USED IN PUBLICATION WHEN THE TRANSPARENT PRACTICE IS USED
Underreporting null effects	Researchers mislead about the strength of the evidence by not reporting conditions or measures that did not support the hypothesis.	Open materials, in which all study materials are reported publicly	Others can see the full study design and fairly evaluate the strength and consistency of the evidence.	
***p*-hacking**	Researchers try many ways of analyzing their data, so the result is more likely to be a fluke rather than a true, replicable pattern.	Open data, in which full data sets are provided	Others can re-run and confirm the statistical analyses. They can also use the data to test new questions.	

HARKing

- The study reveals an unexpected result, but
  - the researcher writes about the study
  - as if the result had been predicted all along.
- Solution?
  - Preregistration, in which researchers publish the hypothesis and study design before data collection and analysis begin
  - Others have more confidence in the strength of the evidence.

Using small samples



- In a small sample, a few chance values can influence the data set,
    - so the study’s estimate is imprecise and less replicable.
- Note:
    - Although it’s not part of research transparency, larger samples are now required and encouraged.
    - Studies with large sample sizes produce estimates that are more precise and replicable.

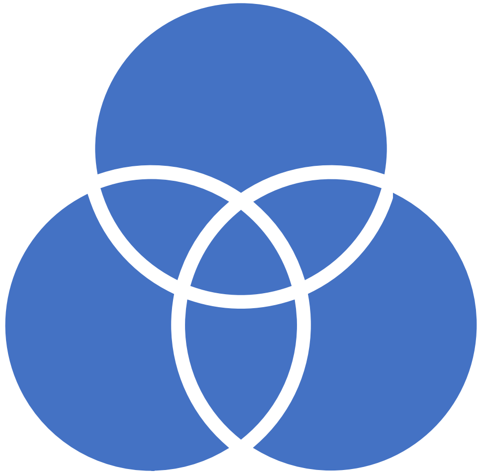
HARKing and Using Small Samples

HARKing	The study reveals an unexpected result, but the researcher writes about the study as if the result had been predicted all along.	Preregistration, in which researchers publish the hypothesis and study design before data collection and analysis begin	Others have more confidence in the strength of the evidence.	
Using small samples	In a small sample, a few chance values can influence the data set, so the study's estimate is imprecise and less replicable.	Although it's not part of research transparency, larger samples are now required and encouraged.	Studies with large sample sizes produce estimates that are more precise and replicable.	

Must a Study Have External Validity?

- Generalizing to other participants
- Generalizing to other settings
- Does a study have to be generalizable to many people?
- Does a study have to take place in a real-world setting?

Generalizing to Other Participants



- It's *a* population, not *the* population.
- External validity comes from *how*, not *how many.*
- Just because a sample comes from a given population doesn't mean that it generalizes to that population.

Generalizing to Other Settings

- **Ecological validity:** An aspect of external validity in which the focus is on whether a laboratory study generalizes to real-world settings

Does a Study Have to Be Generalizable to Many People?

Theory-testing mode

Generalization mode

Theory-Testing Mode

- When in theory-testing mode, external validity and real-world applicability are not as important.
- Examples
  - Contact comfort theory
  - “Parent-as-grammar-coach” theory

Generalization Mode

used when researchers want to generalize the findings from the sample in the study

- to a larger population of interest.

It is important to use probability samples.

The primary concern is external validity.

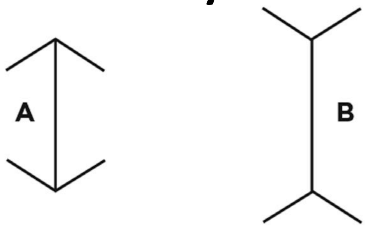
- Applied research tends to be done in generalization mode.
- Some research has aspects of both modes.

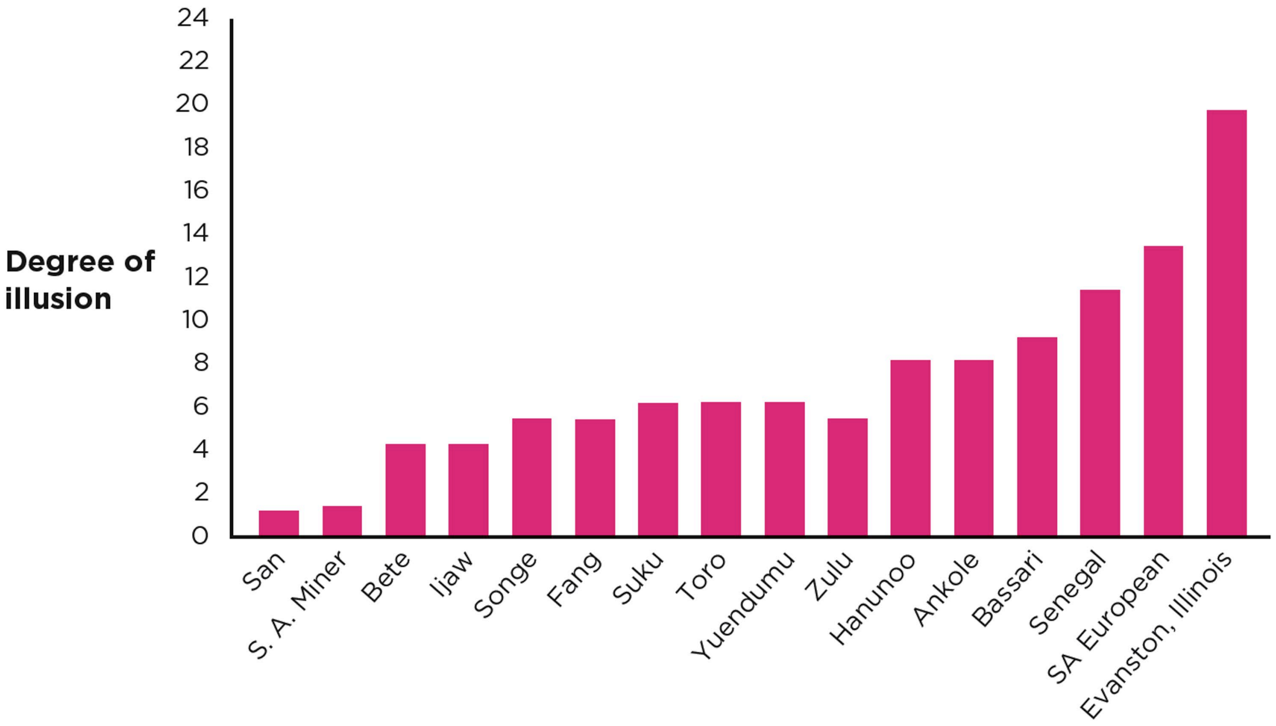
Frequency claims: Always in generalization mode

Association and causal claims: Sometimes in generalization mode

Cultural Psychology: A Special Case of Generalization Mode

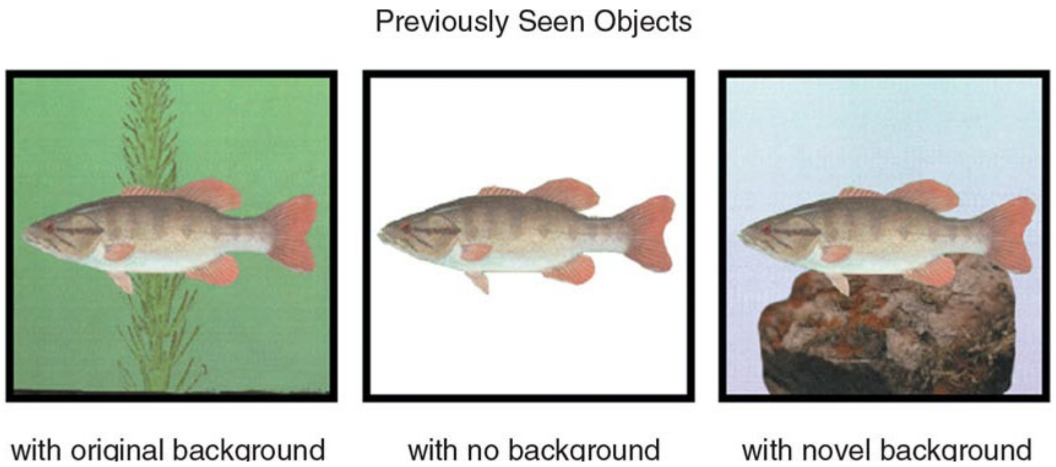
- The Müller-Lyer illusion

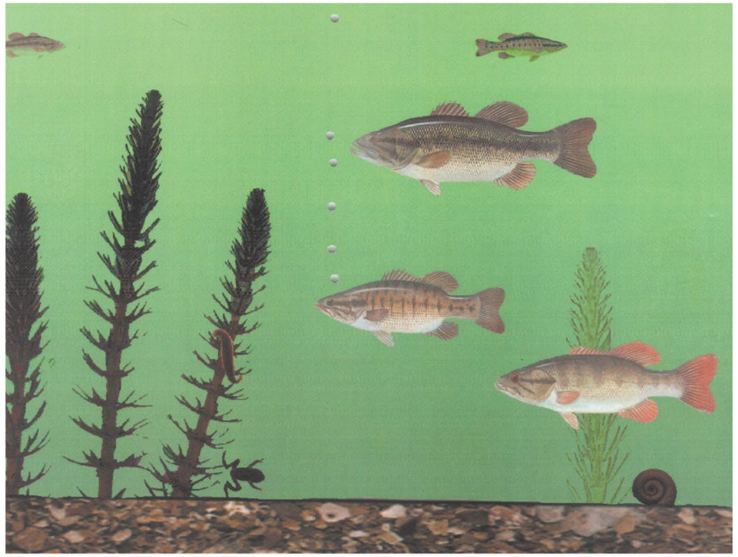




Cultural Psychology: A Special Case of Generalization Mode (Figure and Ground)

Figure and ground





Theory Testing Using WEIRD Participants

Most research in psychology has been conducted on North American college students.

WEIRD cultures = Western, educated, industrialized, rich and democratic

Does a Study Have to Take Place in a Real-World Setting?

External validity and the real world

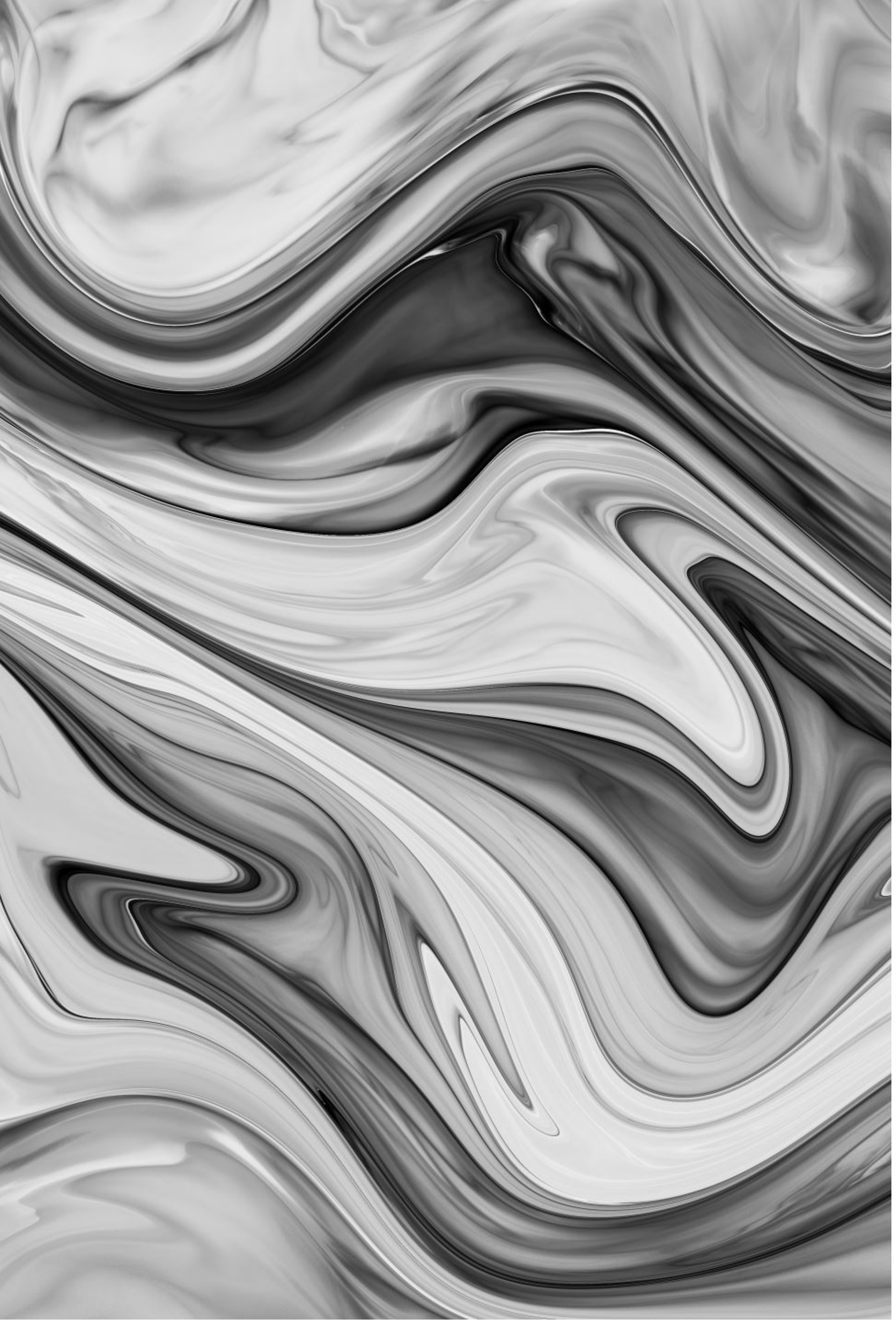
Generalization mode and the real world

Theory-testing mode and the real world

External Validity and the Real World

When a study takes place it the real world, it occurs in a **field setting** and has high external validity.

Laboratory research can be just as realistic as research conducted in the real world, which is known as **experimental realism.**



Generalization Mode and the Real World

- External validity is important when in generalization mode, so having a representative sample is paramount.
- Ecological validity is also important to consider in terms of generalizing to nonlaboratory settings.

Theory-Testing Mode and the Real World

- When in theory-testing mode,
  - external validity and
  - real-world applicability are not as important.

Responding Appropriately to the "Latest Scientific Breakthrough"

SAY THIS	NOT THAT
"Was that finding replicated?" "That is a single study. How does the study fit in with the entire literature?"	"Look at this single, interesting result!"
"How well do the methods of this study get at the theory they were testing?"	"I don't think this study is important because the methods had nothing to do with the real world."
"Was the study conducted in generalization mode? If so, were the participants representative of the population of interest?"	"I reject this study because they didn't use a random sample." "This is a bad study because they used only North Americans as participants."
"The study had thousands of participants, but were they selected in such a way to allow generalization to the population of interest?"	"They used thousands of participants. It must have great external validity."
"Would the result hold up in another cultural context?" "Would that result apply in other settings?"	"That psychological principle seems so basic, I'm sure it's universal."

Wrapping Up…